

Peer Review C3 - Problem 6.3

Given that 2 is a primitive root modulo 101, find $x \in \mathbb{Z}/101\mathbb{Z}$ such that $\text{ord}_{101}(x) = 10$.

Solution

NO AI USED We are given that the modulus is the prime number $p = 101$. We are also given that $g = 2$ is a primitive root modulo 101.

The order of the multiplicative group $\mathbb{Z}/101\mathbb{Z}^\times$ is $\phi(101) = 101 - 1 = 100$. Since g is a primitive root, every element $x \in \mathbb{Z}/101\mathbb{Z}^\times$ can be written as a power of g , i.e., $x = g^i = 2^i$ for some integer i where $0 \leq i < 100$.

The order of an element g^i is given by the formula:

$$\text{ord}_p(g^i) = \frac{p-1}{\gcd(p-1, i)}$$

In our case, we have $p = 101$ and we require $\text{ord}_{101}(x) = 10$. Substituting the known values:

$$10 = \frac{100}{\gcd(100, i)}$$

This implies that $\gcd(100, i)$ must be 10:

$$\gcd(100, i) = \frac{100}{10} = 10$$

We seek an integer i such that $\gcd(100, i) = 10$. Let $i = 10k$ for some integer k . Substituting this:

$$\gcd(100, 10k) = 10$$

Factoring out 10:

$$10 \cdot \gcd(10, k) = 10$$

Dividing by 10:

$$\gcd(10, k) = 1$$

We have to choose an integer k such that k is coprime to 10. The integers that satisfy this condition are $k \in \{1, 3, 7, 9\}$.

Thus, we set $k = 1, 3, 7, 9$, which gives the exponents i :

$$i_1 = 10 \cdot 1 = 10$$

$$i_2 = 10 \cdot 3 = 30$$

$$i_3 = 10 \cdot 7 = 70$$

$$i_4 = 10 \cdot 9 = 90$$

The elements x we are looking for are $x = 2^i$:

$$x = 2^{10} \pmod{101} \equiv 14 \pmod{101}$$

$$x = 2^{30} \pmod{101} \equiv 17 \pmod{101}$$

$$x = 2^{70} \pmod{101} \equiv 6 \pmod{101}$$

$$x = 2^{90} \pmod{101} \equiv 65 \pmod{101}$$

The required values are $\mathbf{x} \in \{\mathbf{6}, \mathbf{14}, \mathbf{17}, \mathbf{65}\}$.